Linear systems and elimination

The exact four-step amplification of restarted Anderson acceleration

Difficulty: challenging **Importance:** interesting to the community

Status: explicit conjecture in the 2025 journal version

Let $n \geq 2$ and let $M \in \mathbb{R}^{n\times n}$ be nonzero and symmetric, with $1 \notin \sigma(M)$. Put $A = I - M$. Define the positively homogeneous map

$$R(v) = M\left(v - \frac{v^T Av}{\|Av\|_2^2} Av\right) \quad (v \neq 0), \qquad R(0) = 0.$$

Two steps of restarted Anderson acceleration with memory one applied to $x = Mx + b$ propagate the residual by R. If $m_1, \ldots, m_n$ are the eigenvalues of M, is

$$\max_{v\neq 0} \frac{\|R(R(v))\|_2}{\|v\|_2} = \max_{i\neq j} \left(\frac{m_i m_j (m_j - m_i)}{|m_i(m_i - 1)| + |m_j(m_j - 1)|}\right)^2 ?$$

A term with zero denominator is defined as zero: under the assumptions this can occur only when $m_i = m_j = 0$.

The conjecture says that a largest four-step residual amplification is attained using only two orthogonal eigenvectors, irrespective of the dimension. It supplies an exact worst-case factor for this restart scheme, relevant to accelerated stationary solvers and multigrid.

References

O. A. Krzysik, H. De Sterck, and A. Smith, *Asymptotic convergence of restarted Anderson acceleration for certain normal linear systems*, SISC 47(2025), Conjecture 10, (9),(16),(24),(27) (journal; arXiv v4). Conjecture 10 states the maximum through two-eigenvector nonlinear eigenvalues; equation(24) evaluates their scalar maximum. The map formulation above also covers exact termination without an undefined $\alpha(0)$.

Status check — 2026-09-08

The arXiv history lists v4,12 May 2025, as latest. Its Conjecture 10 and journal abstract retain the conditional result. Searches for the paper title, “restarted Anderson Conjecture 10”, and later proof/counterexample results found no resolution. Windowed AA, GMRES(1), and restarted CG use different residual maps. The September 2026 Forsythe paper resolves the CG restart question and is not a solution claim for the map above.